Implementing a Fuzzy Inference System

```
def tri(x, a, b, c):
    """Triangular membership function."""
    if x <= a or x >= c:
        return 1.0 if (x == b) else 0.0
    return (x - a) / (b - a) if x < b else (c - x) / (c - b)


def fuzzy_tipper(service, food):
    """
    Mamdani FIS. Inputs: service (0-10), food (0-10). Output: tip % (0-25).
    Rules: 1) service poor  OR food rancid   -> tip low
           2) service good                   -> tip medium
           3) service excellent OR food delicious -> tip high
    """
    # 1. Fuzzification
    s_poor, s_good, s_exc = tri(service, -5, 0, 5), tri(service, 0, 5, 10), tri(service, 5, 10,
        15)
    f_rancid, f_deli = tri(food, -5, 0, 5), tri(food, 5, 10, 15)

    # 2. Rule evaluation (OR = max)
    low = max(s_poor, f_rancid)
    med = s_good
    high = max(s_exc, f_deli)

    # 3. Aggregation + 4. Centroid defuzzification
    num = den = 0.0
    x = 0.0
    while x <= 25:
        mu = max(min(low, tri(x, 0, 0, 13)),
                 min(med, tri(x, 0, 13, 25)),
                 min(high, tri(x, 13, 25, 25)))
        num += x * mu
        den += mu
        x += 0.1
    return num / den if den else 0.0, (low, med, high)


def program11():
    print("\n=== 11. FUZZY INFERENCE SYSTEM (Tip calculator) ===")
    for service, food in [(2, 3), (5, 5), (9, 8), (10, 10)]:
        tip, (lo, me, hi) = fuzzy_tipper(service, food)
        print(f"service={service:>2}, food={food:>2} | rule strengths low={lo:.2f} "
              f"med={me:.2f} high={hi:.2f} | tip = {tip:.2f}%")


if __name__ == "__main__":
    program11()
```